

Seongsu Kim

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Ph.D. Student

Graduate School of Artificial Intelligence

 [seongsukim-ml.github.io](https://github.com/seongsukim-ml)

 seongsu.kim@kaist.ac.kr

RESEARCH INTEREST

My research interests focus on building end-to-end AI systems for scientific discovery, spanning **large-scale model development**, **data generation**, **atomistic simulation**, and **scientific validation**. My work has focused on physically grounded modeling of atoms, molecules, and materials, including **solid-state physics** [C1], **molecular science** [C1,C4,C5], **density functional theory (DFT)** [C1,C4,C5], and **structure prediction and generation** for **MOFs** [C2,C3,C6]. I am also interested in extending these ideas toward **molecular dynamics**, **protein and biomolecular design**, and **autonomous laboratories**.

EDUCATION

9/2025 - current **Ph.D. Student, Korea Advanced Institute of Science and Technology (KAIST), Korea**
Graduate School of Artificial Intelligence
Advisor: Sungsoo Ahn

2/2023 - 8/2025 **M.S., Pohang University of Science and Technology (POSTECH), Korea**
Graduate School of Artificial Intelligence
Advisor: Sungsoo Ahn, and Dongwoo Kim

3/2016 - 2/2023 **B.S., Gwangju Institute of Science and Technology (GIST), Korea**
Majored in *Physics*
Minored in *Mathematics, Computer Science, Artificial Intelligence*

7/2017 - 8/2017 **University of California, Berkeley**
Summer study abroad program

PUBLICATIONS

[C] Conference
[W] Workshop
[P] Preprint
† Corresponding

[W1] Spectrally Constrained Electronic Density Matrix Learning with Subspace Alignment
[Seongsu Kim](#), Honghui Kim, Sungsoo Ahn[†], and Manasa Kaniselvan[†]
Sim2Science NeurIPS Workshop, **2026**

[C6] AtomMOF: All-Atom Flow Matching for MOF-Adsorbate Structure Prediction
Nayoung Kim, Honghui Kim, Sihyun Yu, Minkyu Kim, [Seongsu Kim](#), and Sungsoo Ahn[†]
Neural Information Processing Systems (NeurIPS), **2026**, [PDF](#) [CODE](#)

[C5] Machine Learning Hamiltonians are Accurate Energy-Force Predictors
[Seongsu Kim](#), Chanhui Lee, Yoonho Kim, Seongjun Yun, Honghui Kim, Nayoung Kim, Changyong Park, Sehui Han, Sungbin Lim[†], and Sungsoo Ahn[†]
International Conference on Machine Learning (ICML), **2026**, [PDF](#) [CODE](#)

[P1] MADField: Multi-fidelity Amortized Density Field for Adsorption in Nanoporous Materials
Yoonho Kim, [Seongsu Kim](#), Sungsoo Ahn[†], and Honghui Kim[†]
Preprint, **2026**

[C4] High-order Equivariant Flow Matching for Density Functional Theory Hamiltonian Prediction
[Seongsu Kim](#), Nayoung Kim, Dongwoo Kim[†], and Sungsoo Ahn[†]
Neural Information Processing Systems (NeurIPS), **2025**, [PDF](#) [CODE](#) [SLIDE](#)
Spotlight (3.1%≈688/21575)

[C3] Flexible MOF Generation with Torsion-Aware Flow Matching
Nayoung Kim, [Seongsu Kim](#), and Sungsoo Ahn[†]
Neural Information Processing Systems (NeurIPS), **2025**, [PDF](#)

[C2] MOFFlow: Flow Matching for Structure Prediction of Metal-Organic Frameworks
Nayoung Kim, [Seongsu Kim](#), Minsu Kim, Jinkyoo Park, and Sungsoo Ahn[†]
International Conference on Learning Representations (ICLR), **2025**, [PDF](#)
NeurIPS AIDrugX Workshop, **2024**

[C1] Gaussian Plane-Wave Neural Operator for Electron Density Estimation
[Seongsu Kim](#) and Sungsoo Ahn[†]
International Conference on Machine Learning (ICML), **2024**, [PDF](#) [CODE](#) [SLIDE](#)

EXPERIENCE

10/2026 - 12/2026 (Planned)	Computational Nanoelectronics Group, ETH Zürich <i>Department of Information Technology and Electrical Engineering, Switzerland</i> Advisor: Manasa Kaniselman • Machine learning for open-shell electronic structure and quantum transport materials	Visiting Researcher
2/2025 - current 2/2023 - 2/2025	Structure and Probabilistic Machine Learning (SPML) Lab, Korea KAIST , <i>Korea Advanced Institute of Science and Technology</i> (Advisor: Prof. Sungsoo Ahn) POSTECH , <i>Pohang University of Science and Technology</i> • Machine learning for Scientific Research • Project 1: Accelerating the Density Functional Theory • Project 2: Designing Metal-Organic Framework	Student researcher
9/2021 - 2/2023	Computational Many-body Physics (CMBP) Lab, Korea GIST , <i>Gwangju Institute of Science and Technology</i> (Advisor: Prof. Donghee Kim) • Computer-simulated thermodynamics of <i>solid states physics</i> • Investigated the phase transition of physical models using the Monte Carlo method • Investigated the critical phenomena in the 2D long-range antiferromagnetic Ising model with anisotropy • Wrote the simulation code with C++, MPI and CUDA programming	Research Intern
6/2022 - 7/2022	Statistical Artificial Intelligence (SAIL) Lab, Korea KAIST , <i>Korea Advanced Institute of Science and Technology</i> (Advisor: Prof. Jaesik Choi) • Investigated the various techniques of <i>explainable A.I.</i> including LIMEs, LRP, CRP, and GRAD-CAM.	Research Intern
12/2019 - 2/2020	Quantum Field & Gravity Theory Group, Korea GIST , <i>Gwangju Institute of Science and Technology</i> (Advisor: Prof. Keunyoung Kim) • Investigated the correspondence of deep learning and the Ads/CFT	Research Intern

TALKS & PRESENTATION

10/31/2025	QHFlow: Accelerating DFT with Equivariant Flow Matching NVIDIA BioNeMo Team (Hosted by Youhan Lee) SLIDE	Invited Talk
5/6/2025	Accelerating the <i>ab-initio</i> Calculation with the Machine Learning KAIST-MILA Prefrontal AI Research Center	Invited Talk
15/7/2024	Gaussian Plane-wave Neural Operator for Electron Density Estimation KAIST-POSTECH joint AI Workshop	Presentation

HONORS & AWARDS

2026	Google TPU Builders Award Supported by USD 25,000 in Google Cloud credits for TPU-based AI research	Research Credit
2016 - 2023	Government-Sponsored Tuition Scholarship Received scholarship 8 times (\$20,000)	Scholarship
2016 - 2017	Government-Sponsored Presidential Science Scholarship Received scholarship 2 times (\$1,400)	Scholarship

REVIEWER

NeurIPS (2025), **ICLR** (2026, 2025 [Notable reviewer](#)), **ICML** (2026 [Gold reviewer](#), 2025), **AAAI** (2023)

WORK EXPERIENCE

1/2020 - 8/2021	Republic of Korea Army, Korea Mandatory military service
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LANGUAGES

English - Professional Working, **Korean** - Native

SKILLS

Backgrounds	Computational physics, Quantum mechanics, Statistical physics
Languages	Python (Proficient) , C++, C, Java
Python Libraries	PyTorch, Lightning, Hydra, PyG, WandB, Numpy, Scikit-learn, Matplotlib
Softwares, etc.	Version control (Git and GitHub), Linux-based environment, Vim, Slurm, Docker

DFT tools	PySCF, ORCA, VASP, Quantum Espresso, Castep
CSP tools	GULP, USPEX, CrySPY